



Received:

2025/05/12

Accepted:

2025/06/16

Published:

2025/07/01

RESEARCH PAPER

OPEN ACCESS

Diversity and Distribution of Coral Community from Visakhapatnam Coast, Andhra Pradesh

Joseph Stanley Yogesh Kumar^{1*}, Arya Sen¹, Pradip Panda¹
and Chelladurai Raghunathan²

¹Zoological Survey of India, Sunderban Regional Centre, Canning - 743329, WB, India

²Zoological Survey of India, M-Block, New Alipore, Kolkata - 700053, WB, India

*Correspondence for materials should be addressed to JSYK (email: yogeshkumarzsi16@gmail.com)

Abstract

Coral reefs are crucial ecosystems that support marine biodiversity and provide invaluable services to human communities. The Indo-Pacific region boasts the greatest coral reef diversity worldwide, with over 600 species of reef-building corals and 2,000 species of reef fishes. In India, the four primary coral reef regions include the Gulf of Mannar and Palk Bay, the Gulf of Kachchh, the Lakshadweep Archipelago, and the Andaman and Nicobar Islands. Despite their importance, these reefs face significant threats from climate change, ocean acidification, pollution, and overfishing, leading to a decline in coral cover and diversity, which undermines the productivity and resilience of these ecosystems. Information on coral coverage in Andhra Pradesh, however, remains limited. A recent study conducted from 2019 to 2023 surveyed four sites (15 subsites) along the Visakhapatnam coast using SCUBA diving and digital photography to document coral diversity. The findings revealed that Chintapalle Beach exhibited the highest live coral cover at 23%. This research underscores the ecological value of Andhra Pradesh's coral diversity and highlights the urgent need for robust conservation and management efforts to protect these critical marine habitats.

Keywords: Andhra Pradesh; Hexacorallia; Scleractinia; Scuba; Underwater; Visakhapatnam

Introduction

Coral reefs are among the most diverse and productive ecosystems on the planet, providing critical services to both marine life and human communities (Denley et al., 2020). These underwater oases are home to a vast array of marine organisms, including a wide variety of hard and soft corals, fish, invertebrates, and other species (Frias-Torres et al., 2023). Globally, coral reef diversity is highest in the Indo-Pacific region in shallow to deeper water, and their distribution and abundance are influenced by environmental factors, particularly in the Coral Triangle, which encompasses parts of Southeast Asia and the western Pacific Ocean (Chan et al., 2023). This region is known for its exceptional coral and fish diversity, with estimates of over 600 species of reef-building corals and more than 2,000 species of reef fishes (Høegh-Guldberg et al., 2017). In contrast, coral reef diversity in other regions, such as the Caribbean and the western Indian Ocean, is generally lower, with fewer species of corals and fishes (Bachtar et al., 2023). The high diversity of coral reefs is crucial for the overall health and functioning of these ecosystems (Denley et al., 2020). Diverse coral communities provide a wide range of habitats and resources for other marine organisms, supporting complex food webs and contributing to the overall productivity of the reef (Frias-Torres et al., 2023). Additionally, the structural complexity of coral reefs helps to protect coastal communities from the impacts of storms and waves, and they are important sources of food and income for many coastal communities (Cornwall et al., 2022).

India is home to four major coral reef sites: the Gulf of Mannar and Palk Bay, the Gulf of Kachchh, the Lakshadweep Archipelago, and the Andaman and Nicobar Islands. In addition, patch reefs can be found in various locations, including Ratnagiri, Malvan, Redi, Vengurla to Vijaydurg, and Angria



Bank along the Maharashtra coast; Grand Island along the Goa coast; Netrani Island along the Karnataka coast; Gaveshana Bank in Gujarat; Quilon and Vizhinjam along the Kerala coast; and Enayam and Puducherry along the Tamil Nadu coast (De et al., 2017; Laxmilatha et al., 2019). In India, a total of 585 coral species spanning 108 genera and 23 families have been documented. Among these, 523 species from 95 genera and 23 families have been recorded in the Andaman and Nicobar Islands, 169 species from 46 genera and 17 families in the Gulf of Mannar, 165 species from 54 genera and 17 families in the Lakshadweep Islands, and 76 species from 30 genera and 11 families in the Gulf of Kachchh. Additionally, 32 species have been identified at Angria Bank, 20 species in the Malvan Marine Sanctuary along the Maharashtra coast, 29 species from 17 genera along the Kerala and Tamil Nadu coasts of the southwest region, 15 species at Grand Island along the Goa coast, 14 species at Netrani Island along the Karnataka coast, and most recently, 12 species from the Puducherry coast (De et al., 2020).

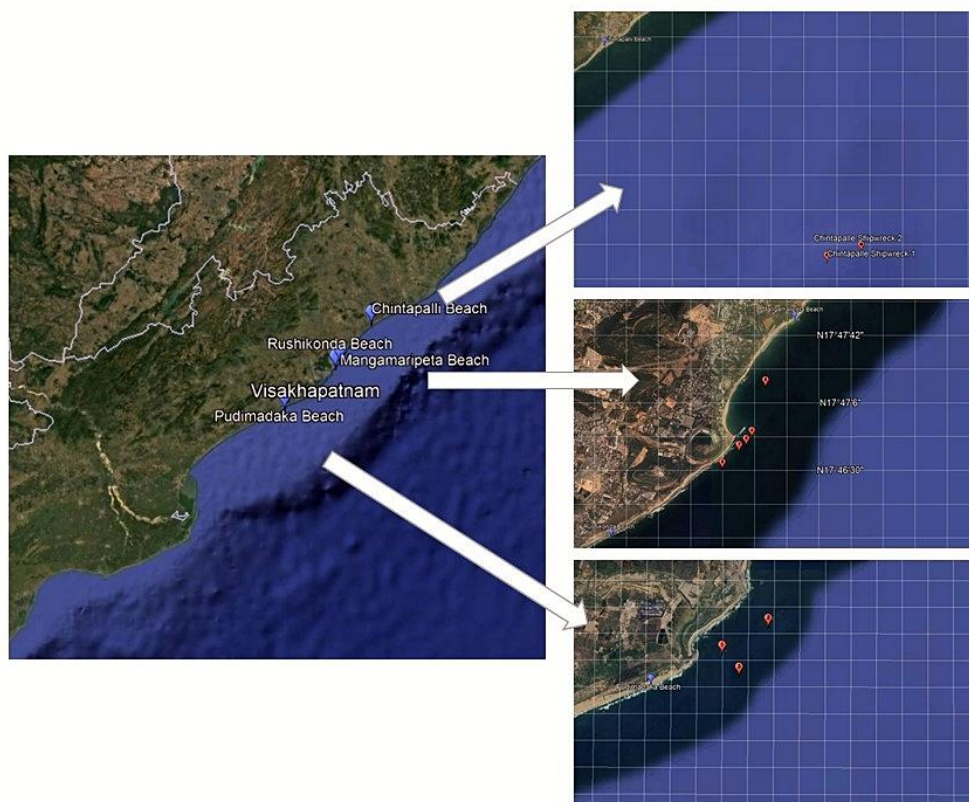


Fig. 1. Study area map for Vishakhapatnam coast, Andhra Pradesh (Pudimadaka beach to Chintapalle beach)

Table 1. Study sites and their co-ordinates with depth

S.No.	Study sites	Co-ordinates	Maximum Depth
1.	Off Pudimadaika Beach	Lat. N 17°28.563' Long. E 82°59.991'	12m
2.		Lat. N 17°28.863' Long. E 83°00.531'	8m
3.		Lat. N 17°28.313' Long. E 83°00.192'	14m
4.	Off Rushikonda beach	Lat. N 17°46.789' Long. E 83°23.364'	9m
5.		Lat. N 17°47.238' Long. E 83°23.503'	6m
6.		Lat. N 17°46.715' Long. E 83°23.319'	9m
7.		Lat. N 17°46.663' Long. E 83°23.251'	8m
8.		Lat. N 17°46.506' Long. E 83°23.094'	9m
9.	Mangamari Peta Beach	Lat. N 17°47.882' Long. E 83°24.296'	8m
10.	Chintapalle Beach	Lat. N 18°00.275' Long. E 83°43.021'	8m
11.		Lat. N 18°00.457' Long. E 83°43.641'	22m
12.		Lat. N 18°00.249' Long. E 83°43.026'	9m
13.		Lat. N 18°00.265' Long. E 83°43.022'	12m
14.		Lat. N 18°00.199' Long. E 83°43.108'	22m
15.		Lat. N 18°00.457' Long. E 83°43.683'	24m

In the past few decades many coral reef surveys have been undertaken by the scientist of Zoological Survey of India (Reddiah, 1970a, b; 1977; Venkataraman and Rajan, 1998; Jeyabaskaran, 1999, 2002; Turner et al., 2001; Venkataraman, 2003; Raghuram and Venkataraman, 2005; Satyanarayana and Ramakrishna, 2009; Rajkumar et al., 2010; Raghuraman et al., 2010; Ramakrishna et al., 2010; Mondal et al., 2010a, b; 2011a, b, c, d, e, f; Venkataraman et al., 2012; Yogesh Kumar and Raghunathan, 2012; Geetha and Yogesh Kumar, 2012; Marimuthu et al., 2013; 2017; 2018; Yogesh

Kumar et al., 2014; 2017a, b, c; 2019). Following the literature review, it can be found that the scattered coral diversity of Andhra Pradesh is under-documented.

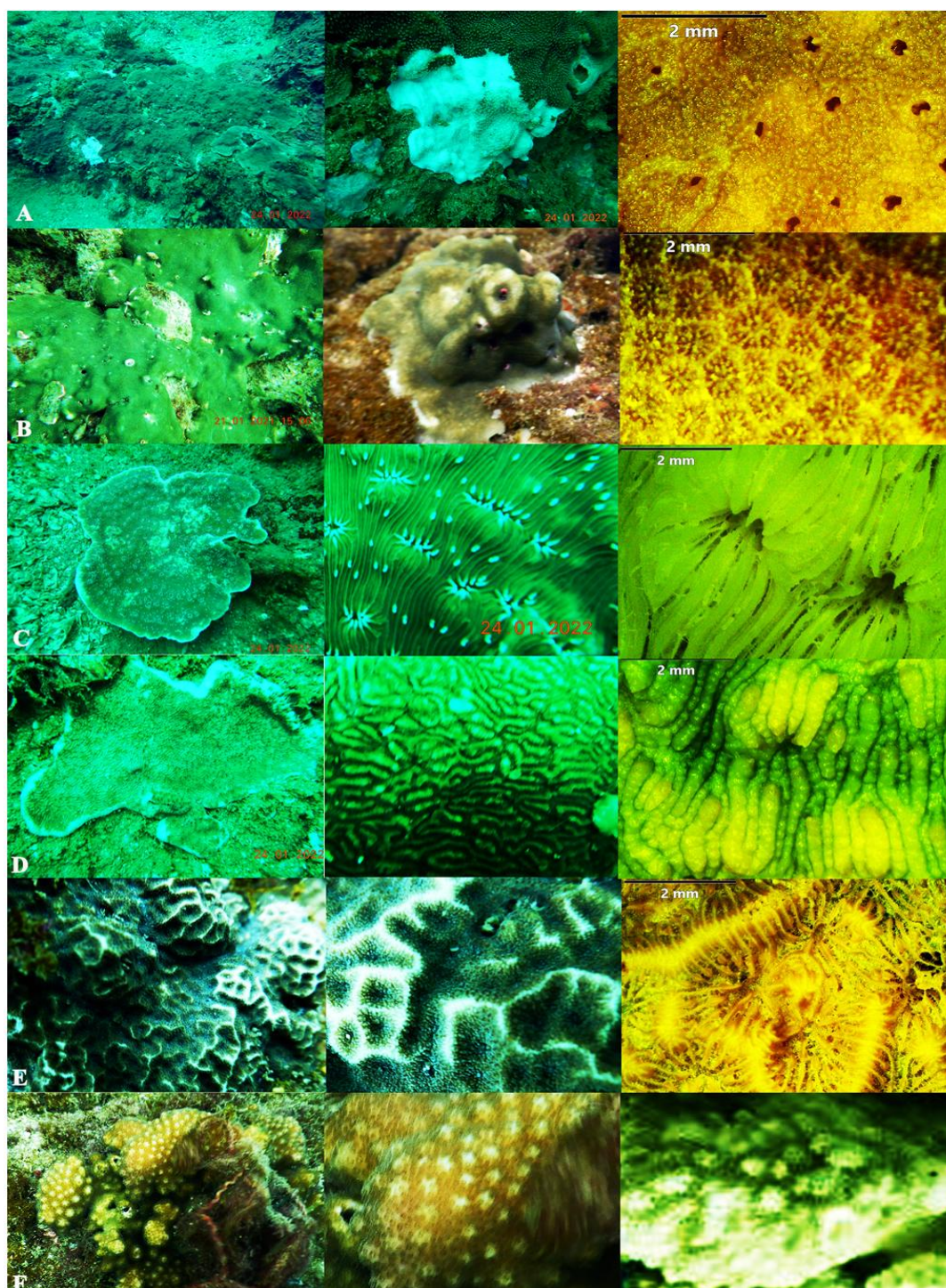


Fig. 2. A- *Porites lichen*, B- *Porites solida*, C- *Leptoseires explanata*, D- *Pavona explanulata*, E- *Pavona venosa*, F- *Pocillopora verrucosa*

Unfortunately, coral reefs around the world are facing a variety of threats, including climate change, ocean acidification, pollution, and overfishing, which are leading to a decline in coral cover and diversity (Samoilys et al., 2022). This loss of coral diversity can have cascading effects on the entire reef ecosystem, leading to a reduction in the overall productivity and resilience of these critical habitats (Gamoyo et al., 2019). Therefore, the present study was undertaken to explore the diversity and distribution of coral reefs in Vishakhapatnam coast, Andhra Pradesh for better conservation efforts.

Methodology

The present study was conducted at 15 sites along the Visakhapatnam coast, Andhra Pradesh, from Pudimadaka to Chintapalle (Table 1, Fig.1), during the period 2019 to 2023. Coral diversity data were collected through SCUBA diving and percentage cover of coral was done using the Coral Video

Transect (CVT) technique (Safuan et al., 2015). The CVT 100 m transect divided into 4 segments with 20 m per segment was overlaid in each site. The coral reef communities were classified into life-form categories, including Live Corals (LC), Bleached Corals (BC), Dead Corals (DC), Algal Assemblages (AA), Sand (SD), Rubble (RR), Rocky Substrates (RK), and Associated Fauna (OT) (English et al., 1997). The percentage cover was estimated, and the collected data were analyzed using biological indices with the PAST software (Hammer, 2012).



Fig. 3. A- *Favia* sp., B- *Tubastrea* sp., C- *Paracyathus stokesii*, D- *Dendrophyllia robusta*, E- *Polycyathus verrilli*, F - *Polycyathus* sp.

Results

The present study covered four major areas along Visakhapatnam coast with a total of 15 sampling sites (Table 1). Sampling sites in Off Pudimadaika Beach were 3, and the depth ranged between 8-14m. Subsequently, a total of 5 sampling sites ranging a depth of 6-9m in Off Rushikonda beach, 1 sampling site with 8m depth range in Mangamari Peta Beach, 6 sampling sites ranging a depth of 8-22m in Chintapalle Beach were covered.

The present study recorded 15 species of corals belonging to 12 genera and 9 families from Visakhapatnam Coast (Table 2). Of which, 12 species were reported from Chintapalle followed by 6 species reported from Rushikonda, 5 species reported from Pudimadaika and 3 species reported from Mangamari Peta. Among them, three coral families were observed very dominantly (Poritidae, Agariciidae, Dendrophylliidae) from 15 study sites (Fig. 2- 4).



Fig. 4. A- *Echinophyllia aspera*, B- *Balanophyllia* sp., C- *Siderastrea savigniana*, D- *Tubastrea* sp., E- *Porites* sp., (Pink band disease) F- *Porites* sp., (Coral bleaching)

In this study maximum percentage of live coral cover was recorded in Chintapalle Beach (23%), followed by Rushikonda beach (20%), Pudimadaika Beach (15%) and Mangamari Peta Beach (5%) (Fig. 5). Two-way Principal Component Analysis (PCA) was carried out between the four sampling sites (Fig. 6). It showed that Rushikonda Beach, Chintapalle Beach stand separate because of the high species diversity and live coral coverage possibly due to the shorter distance and habitat similarity. Pudimadaika Beach and Mangamari Peta Beach were distinct mainly because of the less diversity and different benthic habitat.

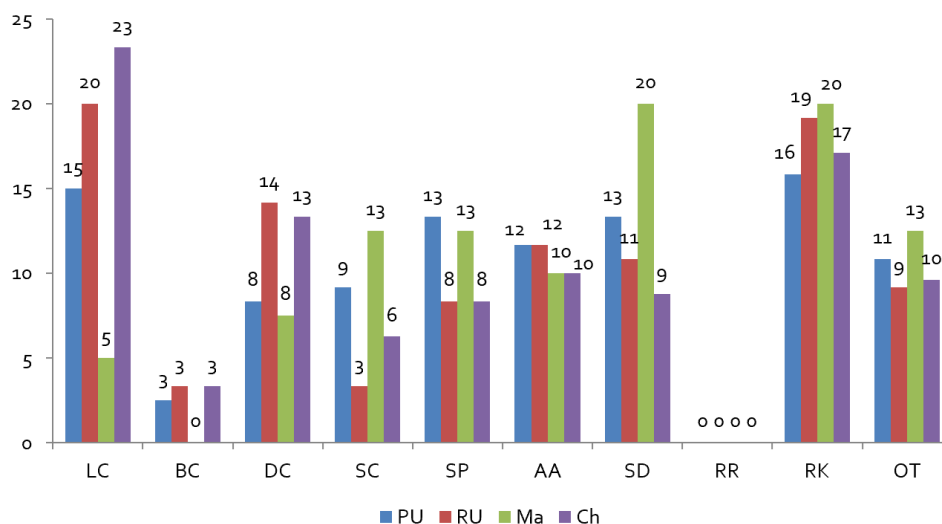


Fig. 5. Percentage of coral cover from the study sites. Pudimadaika Beach (PU); Rushikonda Beach (RU); Mangamari Peta Beach (Ma); Chintapalle Beach (Ch); Live corals (LC); Bleached corals (BC); Dead Corals (DC); Soft corals (SC); Sponges (SP); Algae (AA); Sediment (SD); Rubbles (RR); Rocky (RK); Other (OT)

Table 2. Coral communities from different sampling sites

	Pudimadaika	Rushikonda	Mangamari Peta	Chintapalle
Phylum: Cnidaria Hatschek, 1888				
Class: Hexacorallia Haeckel, 1896				
Order: Scleractinia Bourne, 1900				
Family: Poritidae Gray, 1840				
Genus: <i>Porites</i> Link, 1807				
1. <i>Porites lichen</i> (Dana, 1846)	+	+	+	+
2. <i>Porites solida</i> (Forsk., 1775)	+	+	+	+
Family: Agariciidae Gray, 1847				
Genus: <i>Leptoseris</i> Milne Edwards & Haime, 1849				
3. <i>Leptoseris explanata</i> Yabe & Sugiyama, 1941	—	—	—	+
Genus: <i>Pavona</i> Lamarck, 1801				
4. <i>Pavona explanulata</i> (Lamarck, 1816)	+	+	—	+
5. <i>Pavona venosa</i> (Ehrenberg, 1834)	+	+	—	+
Family: Pocilloporidae Gray, 1840				
Genus: <i>Pocillopora</i> Lamarck, 1816				
6. <i>Pocillopora verrucosa</i> (Ellis & Solander, 1786)	—	—	—	+
Family: Caryophylliidae Dana, 1846				
Genus: <i>Paracyathus</i> Milne Edwards & Haime, 1848				
7. <i>Paracyathus stokesii</i> Milne Edwards & Haime, 1848	—	+	—	+
Family: Dendrophylliidae Gray, 1847				
Genus: <i>Dendrophyllia</i> de Blainville, 1830				
8. <i>Dendrophyllia robusta</i> (Bourne, 1905)	—	—	—	+
Genus: <i>Tubastraea</i> Lesson, 1830				
9. <i>Tubastraea</i> sp.,	+	+	+	+
Genus: <i>Balanophyllia</i> Wood, 1844				
10. <i>Balanophyllia</i> sp.	—	—	—	+
Family: Caryophyllidae				
Genus: <i>Polycyathus</i>				
11. <i>Polycyathus verrilli</i> Duncan, 1889				
12. <i>Polycyathus</i> sp.	+	+	—	+
Family: Lobophylliidae Dai & Horng, 2009				
Genus: <i>Echinophyllia</i> Klunzinger, 1879				
13. <i>Echinophyllia aspera</i> (Ellis & Solander, 1786)	—	—	—	+
Family: Rhizangiidae d'Orbigny, 1851				
Genus: <i>Siderastrea</i> Blainville, 1830				
14. <i>Siderastrea savigniana</i> Milne Edwards & Haime, 1849	+	+	+	+
Family: Faviidae Milne Edwards & Haime, 1849				
Genus: <i>Favia</i> De Blainville, 1820				
15. <i>Favia</i> sp.,	—	—	—	+
Total number of Genera	5	6	3	12

Based on the presence and absence of the species across 4 sampling sites, Bray-Curtis similarity cluster analysis was carried out. For the analysis, the similarity was calculated based on the presence and absence of the species and thus, similarity indices were calculated based on Euclidean distance. Among the 4 sites, the first clusters were formed at a distance of 0.93, Rushikonda Beach, Chintapalle Beach formed the first cluster having almost similar species structure. Both the sites

have higher species diversity and have species from all habitats such as sediment, rubbles, rocky, and other. The second cluster was formed from the first cluster and Pudimadaika Beach and this might be because of the species similarity due to the nearby distance. Mangamari Peta Beach was out clustered which recorded the lowest diversity (Fig. 7).

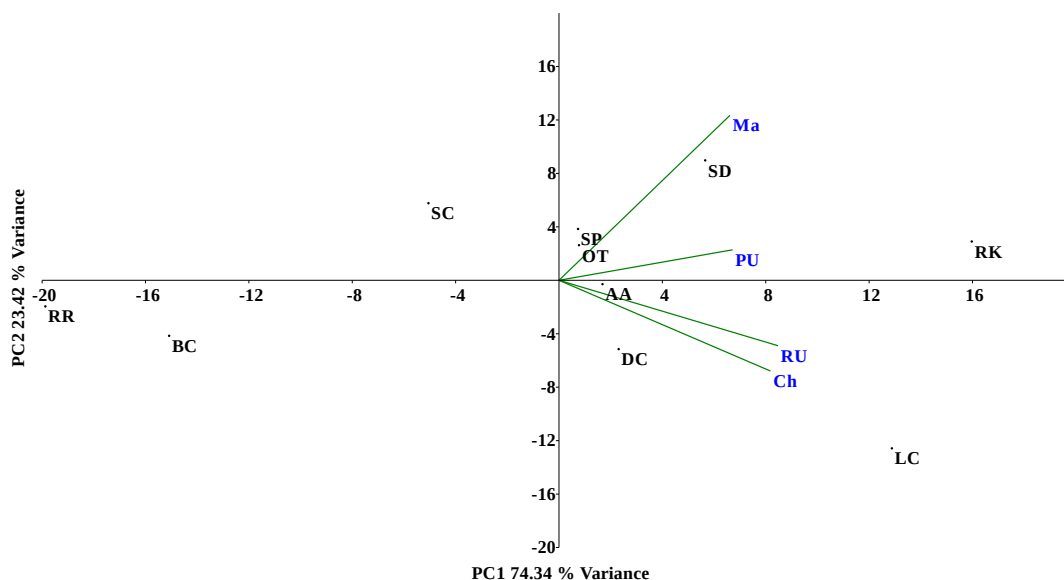


Fig. 6. Principal component analysis based on life-form categories from the study sites. Pudimadaika Beach (PU); Rushikonda Beach (RU); Mangamari Peta Beach (Ma); Chintapalle Beach (Ch); Live corals (LC); Bleached corals (BC); Dead Corals (DC); Soft corals (SC); Sponges (SP); Algae (AA); Sediment (SD); Rubbles (RR); Rocky (RK); Other (OT)

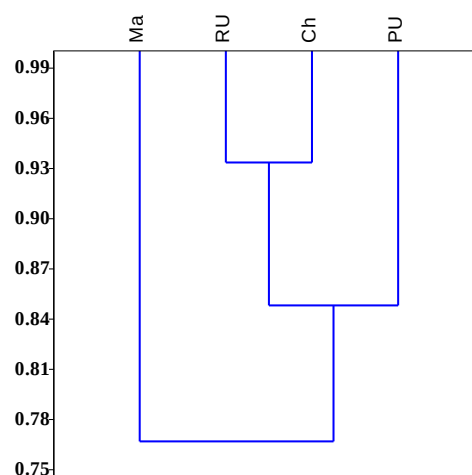


Fig. 7. Bray-Curtis similarity cluster analysis under the paired linked. Pudimadaika Beach (PU); Rushikonda Beach (RU); Mangamari Peta Beach (Ma); Chintapalle Beach (Ch)

Discussion

Coral reefs in India are vital ecosystems that support a rich diversity of marine life, contributing significantly to the ecological and economic well-being of coastal communities. The Indian coral reef system is primarily located in five regions: the Andaman and Nicobar Islands, the Lakshadweep Islands, the Gulf of Mannar, Palk Bay, and the Gulf of Kutch (Jhahhria, 2021). These reefs are predominantly fringing reefs, with the Lakshadweep Islands hosting atolls, which are less common in the Indian context (Jhahhria, 2021). The coral diversity in these regions is influenced by various environmental factors, including water temperature, salinity, and the availability of light, which are crucial for the growth of scleractinian corals (DeVantier and Turak, 2017).

The coral reef diversity along the Visakhapatnam coast of Andhra Pradesh has been the subject of recent investigation, revealing significant insights into the ecological composition and health of these vital marine ecosystems. The study encompassed four major areas, with a total of 15 sampling sites, and highlighted the presence of 15 coral species belonging to 12 genera and 9 families. Notably, the dominant families identified were Poritidae, Agariciidae, and Dendrophylliidae, which are known for their ecological roles in reef-building and habitat complexity Richardson et al. (2017).

Among the sampling sites, Chintapalle Beach exhibited the highest coral diversity, followed by Rushikonda Beach, Pudimadaika Beach, and Mangamari Peta Beach. This variation in species richness can be attributed to several factors, including depth, habitat structure, and local environmental conditions. The maximum percentage of live coral cover was also observed at Chintapalle Beach, indicating a healthier reef ecosystem compared to the other sites, particularly Mangamari Peta Beach, which recorded only 5% live coral cover (Álvarez-Filip et al., 2011). The PCA conducted in the study revealed distinct clusters among the sampling sites, with Rushikonda and Chintapalle Beaches forming a cluster characterized by high species diversity and live coral coverage. This clustering suggests that these sites share similar habitat characteristics, which may facilitate greater coral diversity and resilience (Kandler et al., 2019). Conversely, Pudimadaika and Mangamari Peta Beaches were distinctly separated, likely due to lower species diversity and differing benthic habitats. The Bray-Curtis similarity analysis further supported these findings, indicating that Rushikonda and Chintapalle Beaches have nearly identical species structures, while Mangamari Peta Beach stands out due to its significantly lower diversity (Söffker et al., 2011).

The ecological implications of these findings are profound. High coral diversity is often associated with increased structural complexity, which provides essential habitat for various marine organisms, including fish and invertebrates (Singgalen, 2024). The presence of diverse coral species can enhance the resilience of reef ecosystems against environmental stressors, such as climate change and pollution (Madduppa et al., 2012). However, the observed differences in coral cover and diversity across the sampling sites also highlight the vulnerability of certain areas, particularly Mangamari Peta Beach, which may require targeted conservation efforts to improve its ecological health (Estrada-Saldívar et al., 2019).

Coral reefs in India are not only biodiversity hotspots but also provide essential ecosystem services, including coastal protection, fisheries support, and tourism opportunities (Jhahhria, 2021). The reliance of local communities on coral reef resources underscores the need for effective management and conservation strategies to mitigate the impacts of climate change and human activities. Recent studies have highlighted the importance of coral restoration efforts in regions like Palk Bay, where transplantation of fast-growing branching corals is being implemented to enhance reef recovery (Sadhukhan, 2024). Despite the ecological significance of Indian coral reefs, they are increasingly vulnerable to climate change, leading to coral bleaching events and a decline in coral cover (Pillai, 2024). The health monitoring of coral communities in regions like the Gulf of Mannar has revealed alarming trends, with significant variations in live coral coverage and increased susceptibility to bleaching (Sadhukhan et al., 2022). The ongoing threats necessitate urgent conservation measures, including habitat protection, restoration initiatives, and community engagement in sustainable practices (Jhahhria, 2021).

Conclusion

The coral diversity along the Visakhapatnam coast of Andhra Pradesh underscores the ecological significance of the region and its potential for biodiversity conservation. The present study, which covered four major areas with 15 sampling sites, identified 15 coral species belonging to 12 genera and 9 families. Given the increasing threats to coral reefs globally, including climate change and human impacts, this study reinforces the urgency of implementing effective management strategies. Targeted efforts, such as habitat restoration, monitoring, and community engagement, are essential to safeguard the ecological health and ecosystem services of coral reefs along the Andhra Pradesh coast. Such measures will not only enhance the resilience of these vital marine habitats but also ensure their continued contribution to the ecological and economic well-being of coastal communities.

References

- Álvarez-Filip L, Dulvy NK, Côté IM, Watkinson AR and Gill JA (2011) Coral identity underpins architectural complexity on caribbean reefs. *Ecological Applications* 21(6):2223-2231.
- Bachtiar I, Hadi TA, Karnan K and Bachtiar NT (2023) Other faunas, coral rubbles, and soft coral covers are important predictors of coral reef fish diversity, abundance, and biomass. *Fisheries and Aquatic Sciences* 26(4):268-281.
- Chan YKS, Amri AY, Ang PO, Baria-Rodriguez MV and Chen CA (2023) Decadal stability in coral cover could mask hidden changes on reefs in the east Asian seas. *Communications Biology* 6(1):630.
- Cornwall CE, Carlot J, Branson O, Courtney TA and Harvey B (2022) Increasing importance of crustose coralline algae to coral reef carbonate production under ongoing climate change. *Research Square Preprint*.

- De K, Venkataraman K and Ingole B (2017) Current status and scope of coral reef research in India: A bio-ecological perspective. *Indian Journal of Geo-Marine Sciences* 46(4):647-662.
- De K, Venkataraman K and Ingole B (2020). The hard corals (Scleractinia) of India: A revised checklist. *Indian Journal of Geo-Marine Sciences* 49(10):1651-1660.
- Denley D, Metaas A and Scheibling RE (2020) Subregional variation in cover and diversity of hard coral (scleractinia) in the western province, solomon islands following an unprecedented global bleaching event. *Plos One* 15(11):e0242153.
- DeVantier L and Turak E (2017) Species richness and relative abundance of reef-building corals in the indo-west pacific. *Diversity* 9(3):25.
- English S, Wilkinson C and Baker V (1997) *Survey Manual for Tropical Marine Resources*, 2nd edition. Australian Institute of Marine Sciences, Townsville, Australia.
- Estrada-Saldívar N, Jordán-Dalhgren E, Rodríguez-Martínez RE, Perry CT and Álvarez-Filip L (2019) Functional consequences of the long-term decline of reef-building corals in the caribbean: evidence of across-reef functional convergence. *Royal Society Open Science* 6(10):190298.
- Frias-Torres S, Reveret C, Henri K, Shah N and Montoya-Maya PH (2023) A low-tech method for monitoring survival and growth of coral transplants at a boutique restoration site. *Peer J* 11:e15062.
- Gamoyo M, Obura D and Reason CJC (2019) Estimating connectivity through larval dispersal in the western indian ocean. *Journal of Geophysical Research: Biogeosciences* 124(8):2446-2459.
- Geetha S and Yogesh Kumar JS (2012) Status of corals (Order: Sclerectinia) and associated fauna of Thoothukudi and Vembar group of Islands, Gulf of Mannar, India. *International Journal of Natural Sciences* 3(2):340-349.
- Hammer O (2012) PAST—paleontological statistics version 2.15 refer-ence manual. Natural History Museum, University of Oslo, Norway.
- Høegh-Guldberg O, Poloczanska ES, Skirving W and Dove S (2017) Coral reef ecosystems under climate change and ocean acidification. *Frontiers in Marine Science* 4:158.
- Jeyabaskaran R (1999) Report on repid Zoological Survey of India Andaman and Nicobar Island. Zoological Survey of India, Kolkata.
- Jeyabaskaran R (2002) Diversity of corals and Conservation issues in the Rani Jhansi Marine National Park, Andaman Islands, India. Center of Biodiversity and Conservation, Spring symposium.
- Jhajhria A (2021) Coral reefs of india a review on their status and management. *International Journal of Fisheries and Aquatic Studies* 9(3):206-210.
- Kandler NM, Wooster MK, Leray M, Knowlton N and Voogd NJd (2019) Hyperdiverse macrofauna communities associated with a common sponge, *Stylissa carteri*, shift across ecological gradients in the central red sea. *Diversity* 11(2):18.
- Laxmilatha P, Jasmine S, Sreeram MP and Rengaiyan P (2019) Benthic communities of mesophotic coral ecosystem off Puducherry, east coast of India. *Current Science* 116(6):982-987.
- Madduppa H, Agus SB, Farhan AR, Hh M, Sb A and Ar F (2012) Fish biodiversity in coral reefs and lagoon at the maratua island, east kalimantan. *Biodiversitas Journal of Biological Diversity* 13(3):145-150.
- Marimuthu N, Wilson JJ, Vinithkumar NV and Kirubakaran R (2013) Coral reef recovery status of south Andaman Islands after the bleaching event 2010. *Journal of Ocean University of China* 14(1):91-96.
- Marimuthu N, Verma AK, Yogesh Kumar JS, Adhavan D and Satyanarayana C (2018) Integrated coastal zone management project funded coral restoration process in the Gulf of Kachchh-A present scenario. *Indian Journal of Geo-Marine Sciences* 47(01):15-19.
- Marimuthu N, Yogesh Kumar JS, Raghunathan C, Vinithkumar NV, Kirubakaran R, Sivakumar K and Venkataraman K (2017) North-South gradient of incidence, distribution and variations of coral reef communities in the Andaman and Nicobar Islands, India. *Journal of Coastal Conservation* 21(2):289-301.
- Mondal T, Raghunathan C and Ramakrishna (2010a) New record of nine Scleractinian corals from Rutland Island. *Andaman International Journal of Biological Sciences* 1(3):155-170.

- Mondal T, Raghunathan C and Ramakrishna (2010b) New record of thirteen Scleractinian corals in Indian waters from Middle and North Andaman. *Biosystematica* 4(2):75-89.
- Mondal T, Raghunathan C and Ramakrishna (2011a) Occurrence of seven Scleractinian corals in Ritchie's Archipelago, Andaman Islands of India. *Proceedings of the Zoological Society* 64(1): 57-61.
- Mondal T, Raghunathan C and Ramakrishna (2011b) New record of five Scleractinian corals from Rutland Island, South Andaman Archipelago. *Asian Journal of Experimental Biological Sciences* 2(1):114-118.
- Mondal T, Raghunathan C and Ramakrishna (2011c) New record of six Scleractinian corals to Indian water from Rani Jhonsi Marine National Park and Adjoining Islands, Andaman and Nicobar Archipelago. *Journal of Natural Sciences* 2(2):321-326.
- Mondal T, Raghunathan C, Ramakrishna (2011d) Addition of thirteen Scleractinian as new to Indian water from Rutland Island, Andaman. *Asian Journal of Experimental Biological Sciences* 2(3):383-390.
- Mondal T, Raghunathan C and Ramakrishna (2011e) Notes on three new records of Scleractinian coral from Andaman Islands. *Journal of Oceanography and Marine Science* 2(5):122-126.
- Mondal T, Raghunathan C, Sivaperuman C and Ramakrishna (2010b) Identification of seven Scleractinian corals from Andaman and Nicobar Islands as new record to Indian water. *Proceedings of the Zoological Society* 63(1):61-66.
- Mondal T, Raghunathan C and Venkataraman K (2011f) Five Scleractinian corals as a new record from Andaman Islands – A new addition to Indian marine fauna. *World Journal of Fish and Marine Sciences* 3(5):450-458.
- Pillai RR, Lakshmanan S, Mayakrishnan M, George G and Menon N (2024) Impact of marine debris on coral reef ecosystem of Palk bay, Indian ocean. *Aquatic Conservation: Marine and Freshwater Ecosystems* 34(5):e4160.
- Raghuram KB and Venkataraman K (2005) New record of *Porites annae* crossland and *Porites cylindrical* Dana from Gulf of Mannar and Andaman Water. *Records of the Zoological Survey of India* 105(1-2):133-138.
- Raghuraman R, Sreeraj C and Raghunathan C (2010) Intensive study on the Scleractinian coral diversity of Pongibalu, South Andaman J *Environmental Sociobiology* 7(1):29-36.
- Rajkumar R, Raghuraman R and Satyanarayana C (2010) New records of Scleractinians from Andaman Islands. *Records of the Zoological Survey of India* 110(3):77-92.
- Ramakrishnan ATK, Mondal T, Raghunathan C, Raghuraman R and Sivaperuman C (2010) New record of Scleractinian corals in Andaman and Nicobar Islands. *Records of the Zoological Survey of India Occasional paper* 321:1-144.
- Reddiah K (1970a) The topography of Appa Island and its fringing reef in the Gulf of Mannar. *Indian Science Congress Association Proceedings* 57(4):405-406.
- Reddiah K (1970b) the formation of secondary rock on a reef flat and its effect on reef organisms. *Proceedings of International Symposium on Spiuncula da Echiura Kotor* 1:18-25.
- Reddiah K (1977) The coral reefs of Andaman and Nicobar Islands. *Records of the Zoological Survey of India* 72:315-324.
- Richardson LE, Graham NAJ and Hoey AS (2017) Cross-scale habitat structure driven by coral species composition on tropical reefs. *Scientific Reports* 7(1):7557.
- Sadhukhan K, Ramesh C, Shunmugaraj T and Murthy MR (2022) Health monitoring status of corals in gulf of mannar and palk bay (india) with special reference to coral bleaching event. *Indo Pacific Journal of Ocean Life* 6(2):61-73.
- Sadhukhan K, Shanmuagaraj T and Ramana Murthy MV (2024) Coral restoration in Palk bay on the southeast coast of India. *Research Square preprint*.
- Safuan M, Boo WH, Siang HY, Chark LH and Bachok Z (2015) Optimization of coral video transect technique for coral reef survey: comparison with intercept transect technique. *Open Journal of Marine Science* 5(04):379-397.
- Samoilys M, Álvarez-Filip L, Myers RE and Chabanet P (2022) Diversity of coral reef fishes in the western Indian ocean: implications for conservation. *Diversity* 14(2):102.

- Satyanarayana C and Ramakrishna (2009) Handbook on hard corals of Gulf of Kachchh. Zoological Survey of India, Kolkata, India.
- Singgalen YA (2024) Coral database and monitoring system design for ecological sustainability. *Journal of Information Systems and Informatics* 6(1):270-289.
- Söffker M, Sloman KA and Hall-Spencer JM (2011) In situ observations of fish associated with coral reefs off Ireland. *Deep-Sea Research Part I: Oceanographic Research Papers* 58(8):818-825.
- Turner JR, Vousden D, Klaus R, Satyanarayana C, Fenner D, Venkataraman K, Ralan PT and Subba Rao NV (2001) Report of phase 1: Remote sensing and Rapid site Assessment survey, April 2001. Coral Reef ecosystems of the Andaman Island. Government of India and UNDP, Global Environment Facility.
- Venkataraman K and Rajan PT (1998) Coral reef of Mahatma Gandhi Marine National Park and crown of thorn phenomenon. In: Gangwar B, Chandra K (eds) symposium Proceedings Islands ecosystem and sustainable development, Andaman Science Association and Department of Science and Technology, A&N Administration, Port Blair, pp 124-132.
- Venkataraman K, Raghunathan C, Raghuraman R, Sivaperuman C, Sreeraj CR, Titus I and Yogesh Kumar JS (2012) Scleractinia of Andaman and Nicobar Islands. *Records of the Zoological Survey of India* 320:1- 304.
- Venkataraman K, Satyanarayana CH, Alfred JRB and Wolstenholme J (2003) Handbook on Hard corals of India. Director, Zoological Survey of India, Kolkata.
- Yogesh Kumar JS, Venkatraman C, Geetha S and Sornaraj R (2019) A report on rare occurrence of a mushroom coral *Cycloseris cyclolites* (Lamarck, 1801) in Gulf of Mannar, India. *Indian Journal of Geo-Marine Sciences* 48(3):333-334.
- Yogesh Kumar JS and Raghunathan C (2012) Recovery status of scleractinian corals and associated fauna in the Andaman and Nicobar islands. *Phuket Marine Biological Center Research Bulletin* 71:63-70.
- Yogesh Kumar JS, Satyanarayana Ch and Venkataraman K (2017c) A new scleractinian coral *Lobophyllia hemprichii* (Family: Mussidae) reported first time from the Marine National Park, Gulf of Kachchh, India. *Indian Journal of Geo-Marine Sciences* 46(4):738-741.
- Yogesh Kumar JS, Satyanarayana Ch, Venkataraman K and Chandra K (2017a) Studies on survival and growth rate of transplanted Acroporidae in Gulf of Kachchh Marine National Park, India. *Journal of Coastal Conservation* 21(1):23-34.
- Yogesh Kumar JS, Satyanarayana Ch, Venkataraman K, Beleem IB, Arun G, Chandran R, Ramkumaran K and Kamboj RD (2017b) Coral reefs transplantation and restoration experience in Pirotan Island, Marine National Park, Gulf of Kachchh, India. *Indian Journal of Geo-Marine Sciences* 46(2):299-303.
- Yogesh Kumar JSY, Marimuthu N, Geetha S, Satyanarayana CH, Venkataraman K and Kamboj RD (2014) Longitudinal variations of coral reef features in the Marine National Park, Gulf of Kachchh. *Journal of Coastal Conservation* 18:167-175.

Author Contributions

JSYK, AS, PP and CR conceived the concept, wrote and approved the manuscript.

Acknowledgements

The authors sincerely thank the Director of the Zoological Survey of India, Kolkata, and the Ministry of Environment, Forest, and Climate Change, Government of India, for their invaluable support in providing facilities and financial assistance. They also express their gratitude to the Principal Chief Conservator of Forest (Andhra Pradesh), Chief Conservator of Forest (Visakhapatnam), and Divisional Forest Officer (Vizianagaram) for their support and for granting permission for this study. The authors also deeply appreciate the Scuba support provided by Mr. Balaram Naidu, PADI SCUBA Diving Instructor at Live in Adventures, Rushikonda and Visakhapatnam.

Funding

Not applicable.

Availability of data and materials

Not applicable.

Competing interest

The authors declare no competing interests.

Ethics approval

Consent was obtained from the Director, Zoological Survey of India, India for conducting a survey for better knowledge of the diversity, as well as preparation of the faunal catalogue of the state of Andhra Pradesh.



Open Access *This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution, and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons license, and indicate if changes were made. The images or other third-party material in this article are included in the article's Creative Commons license unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons license and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. Visit for more details <http://creativecommons.org/licenses/by/4.0/>.*

Citation: Yogesh Kumar JS, Sen A, Panda P and Raghunathan C (2025) Diversity and Distribution of Coral Community from Visakhapatnam Coast, Andhra Pradesh. Environmental Science Archives 4(2): 416-427.